

The Black-Scholes Model

(3. Derivation of PDE)

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

The Black Scholes Model

Consider a world with two securities ;

- ▶ a risky asset which is the stock(S) and
- ▶ a risk-free asset which is the bank account(B) which grows as risk-free rate.

Black Scholes assumed the following ✓

- ▶ The risk-free interest rate is known and is constant over time.
- ▶ The asset price follows a geometric Brownian motion with constant volatility.
- ▶ The asset pays no dividends during the life of the option.
- ▶ The option is European-style, which can only be exercised at expiry date.
- ▶ No transaction costs are associated with buying or selling the asset/option.
- ▶ Trading of the asset can take place continuously.
- ▶ Short selling is permitted. ✓

The Delta Hedged Portfolio

Consider a portfolio Π consisting of long Δ unit of Stock(S) and short 1 unit of equity derivative $V(S, t)$.
The value of the portfolio at any time t is given as

$$\Pi = \Delta S - V$$

The Stock follows GBM process $\frac{dS}{S} = \mu dt + \sigma dW$

There is also a bank account that grows at risk-free rate r .

What happens to the portfolio at time $(t + dt)$?

Taylor Series expansion

The change in value of the portfolio in time (dt) is given as

$$d\Pi = \Delta dS - dV$$

Let's expand the right-hand side using Taylor Series expansion.

$$dV(S, t) = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} (dS)^2$$

$$\Rightarrow dV(S, t) = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} dt$$

Substituting these in the above equation

$$d\Pi = \left(\Delta - \frac{\partial V}{\partial S} \right) dS - \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt$$



The No Arbitrage Argument

If the portfolio is risk-free, it will earn the risk-free rate r . This is to prevent no arbitrage between this portfolio and the bank account given both are risk-free. We make the portfolio risk-free by choosing Δ in way that the portfolio is immune to change in stock price (hence no risk).

$$\Delta = \frac{\partial V}{\partial S}$$

With this condition the portfolio will grow at risk-free rate r . In other words,

$$d\Pi = \Pi r dt$$

Substituting in these in the equation

$$\Pi r dt = - \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt$$



The No Arbitrage Argument

Since $\Pi = \Delta S - V$, substituting it in the equation, we get

$$(\Delta S - V)r = -\left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}\right)$$

$$\left(\frac{\partial V}{\partial S} S - V\right)r = -\left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}\right)$$

With a little re-arrangement

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0 \quad (\text{Black-Scholes PDE})$$

Exercise

Show that if the stock pays continuous dividend D , then the PDE is given by

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (\underbrace{r - D}_{\text{circled } r}) S \frac{\partial V}{\partial S} - rV = 0$$

$(dt) \quad - S D dt$

Hint : use this dynamics for Stock $\frac{dS}{S} = (\mu - D)dt + \sigma dW \rightarrow$

Note : For the European option pricing formula, use $S_0 \exp(-DT)$ instead of S_0 . This is because the terminal distribution of S_T is same under

$$\left\{ \begin{array}{l} S_0 + \text{paying dividend yield } D \\ S_0 \exp(-DT) + \text{no dividend yield} \end{array} \right.$$



Yet Another derivation

We superimpose the concept of self-financing strategy to derive the Black Scholes PDE in a very elegant way. So, the derivative at any point is seen as a combination Stock and bank account.

$$V(S, t) = \phi_t S_t + \psi_t B_t$$

Which means $dV(S, t) = \phi_t dS_t + \psi_t dB_t = \phi_t dS_t + \psi_t B_t r dt$

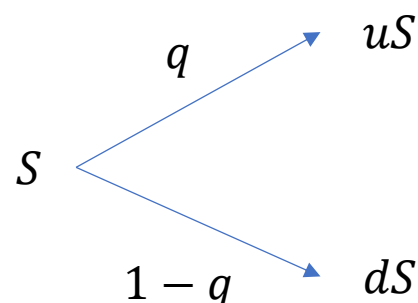
From Ito's lemma we also have $dV(S, t) = \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \frac{\partial V}{\partial S} dS$

Comparing, $\phi_t = \frac{\partial V}{\partial S}$ $\psi_t B_t r = \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right)$

Substituting in the first equation $\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$

BSM PDE from Binomial Tree ?

Can we derive the Black Scholes equation starting from a Binomial Tree set up and then increasing the number of steps infinitely? First, we need to calibrate our binomial tree that matches the stock price dynamics under risk-neutral density in the time interval δt . There are different choices available. The most famous one used in practice is called CRR (Cox-Ross-Rubinstein) tree set up as given below.



$$u = \exp(\sigma\sqrt{\delta t})$$

$$d = \exp(-\sigma\sqrt{\delta t})$$

$$ud = 1$$

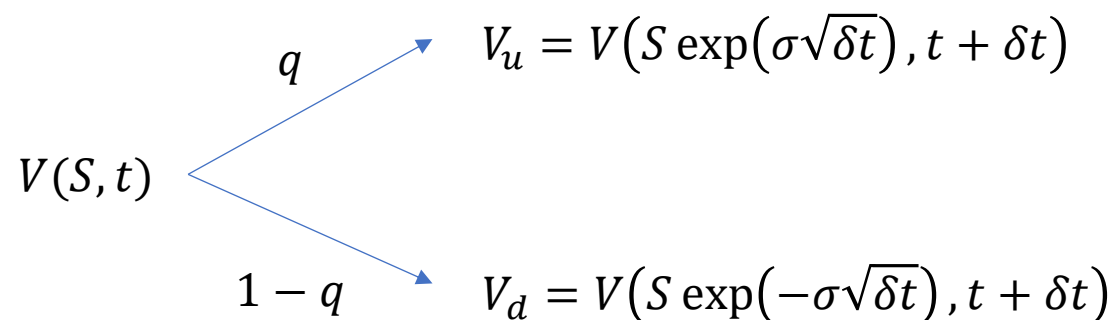
$$q = \frac{\exp(r\delta t) - d}{u - d}$$

Since
$$S_{t+\delta t} = S_t \exp \left[\left(r - \frac{1}{2} \sigma^2 \right) \delta t + \sigma Z \sqrt{\delta t} \right]$$

Verify that
$$\mathbb{E} \left(\ln \left(\frac{S_{t+\delta t}}{S_t} \right) \right) = \left(r - \frac{1}{2} \sigma^2 \right) \delta t \quad \text{and} \quad \mathbb{V} \left(\ln \left(\frac{S_{t+\delta t}}{S_t} \right) \right) = \sigma^2 \delta t \quad \text{on the tree.}$$

BSM PDE from Binomial Tree ?

Now Consider the value of the derivative $V(S, t)$



Using Taylor Series expansion,

$$V_u \approx V(S, t) + \frac{\partial V}{\partial S} S \left(\sigma\sqrt{\delta t} + \frac{1}{2} \sigma^2 \delta t \right) + \left(\frac{\partial V}{\partial t} + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} \sigma^2 S^2 \right) \delta t$$

$$V_d \approx V(S, t) + \frac{\partial V}{\partial S} S \left(-\sigma\sqrt{\delta t} + \frac{1}{2} \sigma^2 \delta t \right) + \left(\frac{\partial V}{\partial t} + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} \sigma^2 S^2 \right) \delta t$$

BSM PDE from Binomial Tree ?

Two equations emerge

Using the result $\Delta = \frac{V_u - V_d}{S_u - S_d}$, we get $\Delta = \frac{\partial V}{\partial S}$

$$y^2 = 4ax$$

$$ax^2 + by^2 + cxy$$

And the pricing equation $V = e^{-r\delta t}(qV_u + (1 - q)V_d)$ leads to

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

Observations

- ▶ The Black-Scholes equation is a **linear parabolic partial differential equation**
- ▶ The equation contains all the key variables such as dependence on underlying (S), time (t), volatility (σ) but there is no mention of the real-world drift μ . σ
- ▶ This means if two people disagree on the drift but agree on the volatility of the asset, then they would agree on the price of the derivative.
- ▶ The derivation of PDE did not assume any specific payoff of the derivative. Note that the PDE is trivially satisfied by both Stock(S) and the Bank account(B). Verify the same.

$S = V$ $\rightarrow \frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$ $\rightarrow V = S$

$0 + (0) + rS - rS = 0$

$V = B$
 $V_t = B_0 e^{rt}$, $B_0 e^{rt} \cdot r + 0 + 0 - r B_0 e^{rt} = 0$

$V(S, t)$ S

$\rightarrow F = f(S)$

$ds/s = \mu dt + \sigma dw$

$df = \frac{df}{ds} ds + \frac{1}{2} \frac{d^2 f}{ds^2} (ds)^2$

Each term in BSM equation represents a critical component in the overall dynamics of the derivative

$$ds/s = (r-D)dt + \sigma dw$$

$$\underbrace{\frac{\partial V}{\partial t}}_{\text{(time decay)}} + \underbrace{\frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}}_{\text{(diffusion)}} + \underbrace{rS \frac{\partial V}{\partial S}}_{\text{(convection/drift)}} - \underbrace{rV}_{\text{(discounting)}} = 0$$

$$\frac{\partial u}{\partial t} = \frac{1}{2}k \frac{\partial^2 u}{\partial x^2}$$

Criticisms of Black Scholes

Black-Scholes equation is grounded in sound economic theory, and it gives analytical price for European options. However, the model suffers from a few practical challenges. All the original assumptions do not necessarily hold true in real life. There are a few other pitfalls worth discussing

- ▶ **Assumption** - Stock price follows lognormal diffusion and returns follow normal distribution. In practice, however, returns are fat tailed (leptokurtic) and negative skewed.
- ▶ **Assumption** – Constant volatility. Volatility is often seen to a function of time and level of the price. More advanced models which incorporate stochastic nature of volatility are used in practice.



Solving the PDE

To arrive at an explicit solution for the PDE, we need to impose the boundary condition which is the payoff of the European security in context. For example, the following boundary conditions apply

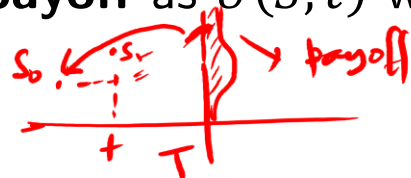
$$V(\underline{S_T}, T) = \begin{cases} \max\{S_T - K, 0\} & \text{if Call} \\ \max\{K - S_T, 0\} & \text{if Put} \end{cases}$$

(Handwritten red annotations: arrows pointing to S_T and T in the first case, and S_T in the second case; checkmarks next to 'Call' and 'Put')'

The Black-Scholes PDE is reducible to the 1-D heat equation using a series of substitutions. Let's review the steps.

Step – 1 (The Expected Payoff process)

At any time t , the value of the derivative is the discounted expected payoff. Let's denote the **expected payoff** as $U(S, t)$ which is related to the price $V(S, t)$ as per the equation below



(+)

$$V(S, t) = e^{-r(T-t)} U(S, t)$$

Note that
$$\frac{\partial V}{\partial t} = e^{-r(T-t)} \frac{\partial U}{\partial t} + r U e^{-r(T-t)}$$

Substituting in the original PDE, we get

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + r S \frac{\partial V}{\partial S} - r V = 0 \quad \longrightarrow \quad \frac{\partial U}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 U}{\partial S^2} + r S \frac{\partial U}{\partial S} = 0$$

Note that the discounting factor disappeared!

Step – 2 (Turn back the time)

Instead of looking at evolution of time t , we can look at time to maturity τ . When t increases τ falls

$t \rightarrow$
 $\tau \rightarrow$

$\tau = T - t$

$\frac{\partial U}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 U}{\partial S^2} + rS \frac{\partial U}{\partial S} = 0$

$\frac{\partial U}{\partial \tau} = \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 U}{\partial S^2} + rS \frac{\partial U}{\partial S}$

$(-1) \cdot \frac{\partial U}{\partial \tau} = \frac{\partial U}{\partial t} \cdot \frac{\partial t}{\partial \tau} = \frac{\partial U}{\partial t} \cdot (-1)$

Step – 3 (Space transformation $S \rightarrow \log S$)

We have performed the log transformation of stock price during derivation of closed form formula for geometric Brownian motion process. In that spirit, let's consider another variable

$$\frac{\partial U}{\partial S} = \frac{\partial U}{\partial \tilde{S}} \times \frac{\partial \tilde{S}}{\partial S} = \frac{\partial U}{\partial \tilde{S}} \times \frac{1}{S}$$

Handwritten notes: $\frac{d(\ln S)}{dS} = \frac{1}{S}$, $\tilde{S} = \log S$

Substituting in the original equation

$$\frac{\partial U}{\partial \tau} = \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 U}{\partial S^2} + rS \frac{\partial U}{\partial S}$$

→

$$\frac{\partial U}{\partial \tau} = \frac{1}{2} \sigma^2 \frac{\partial^2 U}{\partial \tilde{S}^2} + \left(r - \frac{1}{2} \sigma^2 \right) \frac{\partial U}{\partial \tilde{S}}$$

Step – 4 (Removing the Drift)

This next step, we will use the following substitution to remove the drift term.

Substituting in the original equation

$$x = \tilde{S} + \left(r - \frac{1}{2}\sigma^2\right)\tau \quad \text{and} \quad U = W(x, \tau)$$

$$\frac{\partial U}{\partial \tau} = \frac{1}{2}\sigma^2 \frac{\partial^2 U}{\partial \tilde{S}^2} + \left(r - \frac{1}{2}\sigma^2\right) \frac{\partial U}{\partial \tilde{S}}$$

$$\frac{\partial W}{\partial \tau} = \frac{1}{2}\sigma^2 \frac{\partial^2 W}{\partial x^2}$$

Handwritten notes and derivations:

- $v(\tilde{S}, \tau) \rightarrow W(x, \tau)$
- $\frac{\partial U}{\partial \tau} = \frac{\partial W}{\partial \tau}$
- $\frac{\partial^2 U}{\partial \tilde{S}^2} = \frac{\partial^2 W}{\partial x^2}$
- $\frac{\partial U}{\partial \tilde{S}} = \frac{\partial W}{\partial x}$
- $\frac{\partial W}{\partial \tau} = \frac{\partial W}{\partial \tau} + \frac{\partial W}{\partial x} \cdot \frac{\partial x}{\partial \tau} = \frac{\partial W}{\partial \tau} + \frac{\partial W}{\partial x} \cdot \left(r - \frac{1}{2}\sigma^2\right)$
- $\frac{\partial W}{\partial \tau} + \frac{\partial W}{\partial x} \cdot \left(r - \frac{1}{2}\sigma^2\right) = \frac{1}{2}\sigma^2 \frac{\partial^2 W}{\partial x^2}$

The final form is the one dimensional heat equation which we have solved before. The next steps would be to apply the boundary conditions and substitute back all the original variables. This will lead to the final formula for Black-Scholes European call and put options.